Disproof of Conjecture 00000000479

jiakeboge

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The conjecture

Conjecture 00000000479 asserts the following. The perfect matchings (domino tilings) of the Aztec diamond of order n — of which there are $2^{n(n+1)/2}$ — are stratified by the number k of NW–SE main-diagonal lozenges they contain. Then

$$v_2\big(N_k^{(n)}\big) = 0 \text{ if } k \equiv 0 \pmod 4, \qquad v_2\big(N_k^{(n)}\big) = 1 \text{ otherwise,}$$

where $N_k^{(n)}$ is the number of matchings in the k-th stratum and v_2 is the 2-adic valuation. The pattern is asserted to have period 4.

We show this is false already for $n = 1$ and $n = 2$, and that the failure is not an edge phenomenon: it also occurs at $n = 3$ and $n = 4$.

Set-up

For $n \geq 1$ let the Aztec diamond $\mathcal{A}_n$ of order n be the set of unit cells (i, j) — meaning $[i, i+1] \times [j, j+1]$ — with

$$\big|i + \tfrac{1}{2}\big| + \big|j + \tfrac{1}{2}\big| \leq n.$$

Thus $|\mathcal{A}_1| = 4$ and $|\mathcal{A}_2| = 12$, in agreement with $2n(n+1)$ cells. A *domino* is a pair of cells sharing an edge; a *matching* is a set of dominoes partitioning $\mathcal{A}_n$. We call a domino *horizontal* if its two cells have equal second coordinate and *vertical* otherwise.

The conjecture's phrase "NW–SE main-diagonal lozenge" selects one of the two orientations. This choice is immaterial for the stratum sizes: reflection in the line $x = y$ is an involution of $\mathcal{A}_n$ exchanging horizontal and vertical dominoes, so the horizontal and vertical stratum sizes coincide. We therefore count horizontal dominoes; the same numbers hold for vertical ones.

The counterexample

Lemma 1. *For $n = 1$ the two matchings of $\mathcal{A}_1$ carry 0 and 2 horizontal dominoes respectively. Hence $N_2^{(1)} = 1$ and $v_2\big(N_2^{(1)}\big) = 0$.*

Proof. $\mathcal{A}_1$ consists of the four cells $(-1,-1)$, $(-1,0)$, $(0,-1)$, $(0,0)$. There are exactly two matchings: the two horizontal dominoes $\{(-1,-1),(0,-1)\}$ and $\{(-1,0),(0,0)\}$, or the two vertical dominoes $\{(-1,-1),(-1,0)\}$ and $\{(0,-1),(0,0)\}$. □

Since $2 \not\equiv 0 \pmod 4$, the conjecture requires $v_2\big(N_2^{(1)}\big) = 1$, but the value is 0.

Theorem 1. *For $n = 2$ the eight matchings of $\mathcal{A}_2$ have horizontal-domino counts $6, 4, 4, 2, 4, 2, 2, 0$. Consequently*

$$N_0^{(2)} = 1, \quad N_2^{(2)} = 3, \quad N_4^{(2)} = 3, \quad N_6^{(2)} = 1,$$

and in particular $v_2\big(N_2^{(2)}\big) = v_2(3) = 0 \neq 1$ and $v_2\big(N_6^{(2)}\big) = v_2(1) = 0 \neq 1$.

Proof. $\mathcal{A}_2$ has the twelve cells

$$(-2,-1),\ (-2,0),\ (-1,-2),\ (-1,-1),\ (-1,0),\ (-1,1),$$
$$(0,-2),\ (0,-1),\ (0,0),\ (0,1),\ (1,-1),\ (1,0).$$

Enumerating matchings by repeatedly covering the lexicographically least uncovered cell yields exactly the eight listed below, where only the *horizontal* dominoes are written out; every domino not listed is vertical. A pair $(a,b)(c,d)$ denotes the horizontal domino with cells (a,b) and (c,d), and h is the number of horizontal dominoes.

0 $(-2,-1)(-1,-1),\ (-2,0)(-1,0),\ (-1,-2)(0,-2),\ (-1,1)(0,1),\ (0,-1)(1,-1),\ (0,0)(1,0)$; $h = 6$.

1 $(-2,-1)(-1,-1),\ (-2,0)(-1,0),\ (-1,-2)(0,-2),\ (-1,1)(0,1)$; $h = 4$.

2 $(-1,-2)(0,-2),\ (-1,-1)(0,-1),\ (-1,0)(0,0),\ (-1,1)(0,1)$; $h = 4$.

3 $(-1,-2)(0,-2),\ (-1,-1)(0,-1)$; $h = 2$.

4 $(-1,-2)(0,-2),\ (-1,1)(0,1),\ (0,-1)(1,-1),\ (0,0)(1,0)$; $h = 4$.

5 $(-1,-2)(0,-2),\ (-1,1)(0,1)$; $h = 2$.

6 $(-1,0)(0,0),\ (-1,1)(0,1)$; $h = 2$.

7 none; $h = 0$.

Every listed configuration is readily checked to be a matching of $\mathcal{A}_2$, and the eight are distinct. Since the Aztec diamond of order 2 admits exactly $2^{n(n+1)/2} = 2^3 = 8$ matchings, the list is complete. □

Both strata with $k \not\equiv 0 \pmod 4$ have odd cardinality, whereas the conjecture demands 2-adic valuation exactly 1. This contradicts the conjecture.

The failure is not an edge phenomenon

The same enumeration gives the full stratum table:

n	N_0	N_2	N_4	N_6	N_8	N_{10}	$\cdots$
1	1	1					
2	1	3	3	1			
3	1	6	15	20	15	6	$\cdots$
4	1	10	45	120	210	252	$\cdots$

At $n = 3$ we have $N_6^{(3)} = 20$ with $v_2(20) = 2 \neq 1$, and at $n = 4$ the strata $k = 6, 8, 10, 12, 14$ all violate the asserted valuation. For instance $N_8^{(4)} = 210$ is even though $8 \equiv 0 \pmod 4$, so even the "vanishing" half of the dichotomy fails. (Computationally the tables are $N_k^{(n)} = \binom{n(n+1)/2}{k/2}$ for k even, verified for $n \leq 5$; this identity is not needed for the disproof and is recorded only as an observation.)

Formalisation

The Lean 4 project in `lean4/` encodes the order-1 and order-2 Aztec diamonds as explicit finite lists of cells, enumerates their matchings by a total recursive search, and kernel-checks

- that the order-2 diamond admits exactly 8 matchings,
- that exactly 3 of them contain exactly 2 horizontal dominoes, and
- that 3 is odd, so its 2-adic valuation is 0, not 1.

No search is left unformalised: the enumeration is a definition, and the two counting statements are decided in the kernel. No `sorry`, `native_decide`, or additional axiom is used.